A Time Series Analysis of UK Gas Consumption, 1960–1986

Feda Mohammadi

Time Series Econometrics (ECO 402)

Berea College

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Abstract

This paper analyzes the `UKgas` series, which records quarterly gas consumption in the United Kingdom from Q1 1960 through Q4 1986. The series has a clear upward trend, strong quarterly seasonality, and seasonal swings that grow with the level of the series, all of which point to a multiplicative structure. Holt-Winters forecasting and log-linear regression with seasonal dummies both capture the visible deterministic pattern, but residual autocorrelation remains in both cases. After a log transformation and seasonal differencing at lag 4, the series becomes approximately stationary. AR(1) is selected as the preferred model based on AIC and residual diagnostics, with MA(1) a close second and ARMA(1,1) offering no improvement from the added parameter.

1 Dataset Description and Visualization

The dataset used in this paper is the built-in `UKgas` series in R, which records quarterly gas consumption in the United Kingdom from Q1 1960 through Q4 1986. The unit of

measurement is millions of therms. With four observations per year over 27 years, the series contains 108 observations in total. Gas consumption is a useful series to study because it is driven by both long-run factors, such as population growth and industrial expansion, and short-run factors like seasonal temperature changes, which makes it a rich case for time series analysis.

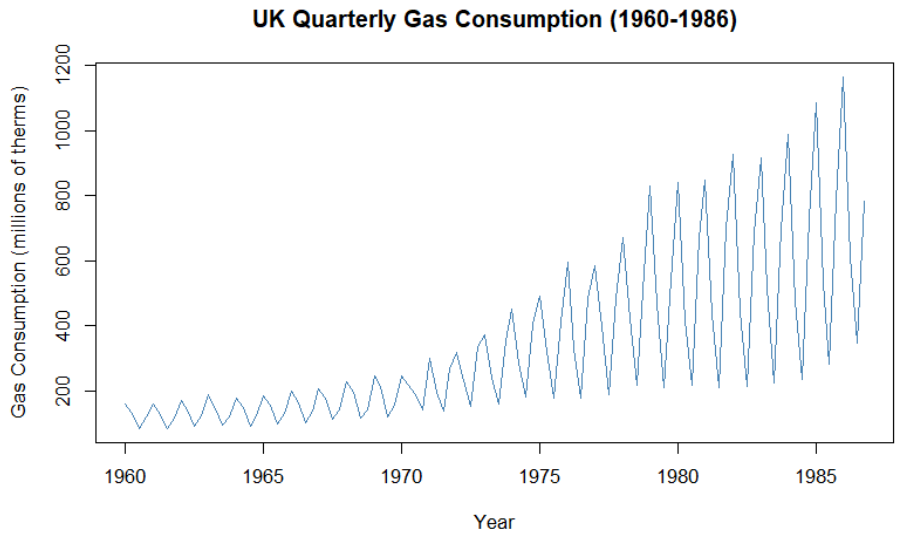


Figure 1: UK Quarterly Gas Consumption (1960–1986)

Figure 1 shows the raw series. Three things we can observe immediately. First, there is a clear upward trend throughout the sample period. Consumption roughly triples from around 160 million therms in Q1 1960 to over 1,100 million therms by Q1 1986. Second, there is strong and very regular quarterly seasonality. Consumption peaks every Q1 (January through March), which corresponds to peak winter heating demand, and drops to its lowest point every Q3 (July through September), when heating is minimal. Third, and importantly, the size of those seasonal swings grows over time as the overall level rises. Early in the sample the gap between Q1 and Q3 is relatively small, but by the 1980s the within-year swings span several hundred million therms. This pattern of growing seasonal variation is a strong visual signal that the series has a multiplicative rather than additive structure, meaning seasonality scales with the trend rather than staying constant.

2 Decomposition and Seasonal Analysis

Because the seasonal swings in the raw series grow visibly with the overall level, a multiplicative decomposition is more appropriate than an additive one. In a multiplicative model, the observed series is expressed as:

$$x_t = T_t \times S_t \times R_t$$

where T_t is the trend component, S_t is the seasonal component, and R_t is the random (irregular) remainder. This structure means that seasonality scales proportionally with the trend, rather than adding a fixed amount on top of it.

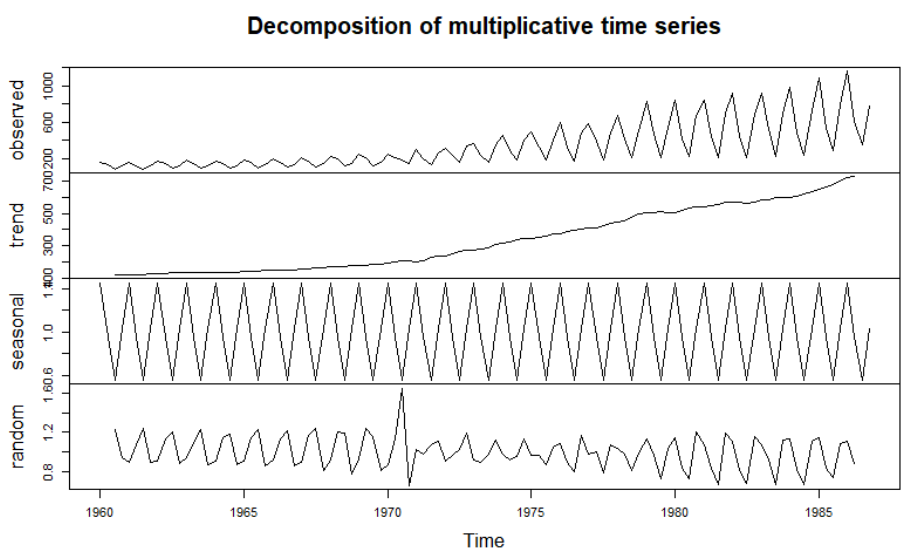


Figure 2: Multiplicative decomposition of the UKgas series

Figure 2 shows the four panels of the decomposition. The trend panel confirms a steady and nearly uninterrupted rise in gas consumption from around 140 million therms in 1960 to over 700 million therms by 1986. The seasonal panel shows a perfectly repeating pattern across all 27 years, which is expected since classical decomposition assigns a single fixed seasonal factor to each quarter. The estimated seasonal factors are 1.454 for Q1, 0.956 for Q2, 0.558 for Q3, and 1.032 for Q4. This means Q1 consumption is on average about 45%

above the trend level, while Q3 consumption is about 44% below it. Q4 is just slightly above the trend and Q2 is just slightly below. The random component fluctuates around 1.0 as expected under multiplicative decomposition, with one noticeable spike around 1970–1971 that likely reflects the disruption caused by the transition from town gas to natural gas in the UK during that period.

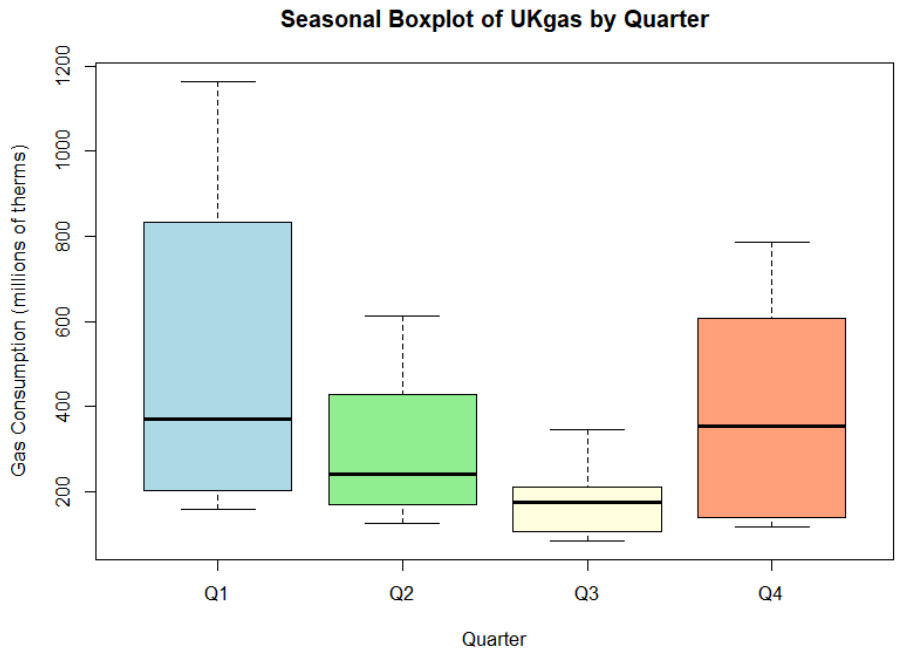


Figure 3: Seasonal boxplot of UKgas by quarter

Figure 3 reinforces the same picture from a different angle. Q1 has by far the highest median consumption and the widest spread, which reflects both the dominance of winter heating demand and the fact that consumption grew substantially over the sample period. Q3 has the lowest median and the tightest spread, consistent with minimal heating needs in summer. Q4 sits higher than Q2, likely because autumn temperatures drop earlier than spring temperatures rise. Taken together, the decomposition and the boxplot confirm that the seasonal pattern in UKgas is strong, regular, and multiplicative.

3 Autocorrelation Analysis

Before fitting any model, it is important to examine the autocorrelation structure of the raw series. The ACF and PACF plots together give a clear picture of the dependence in the data and whether the series is stationary.

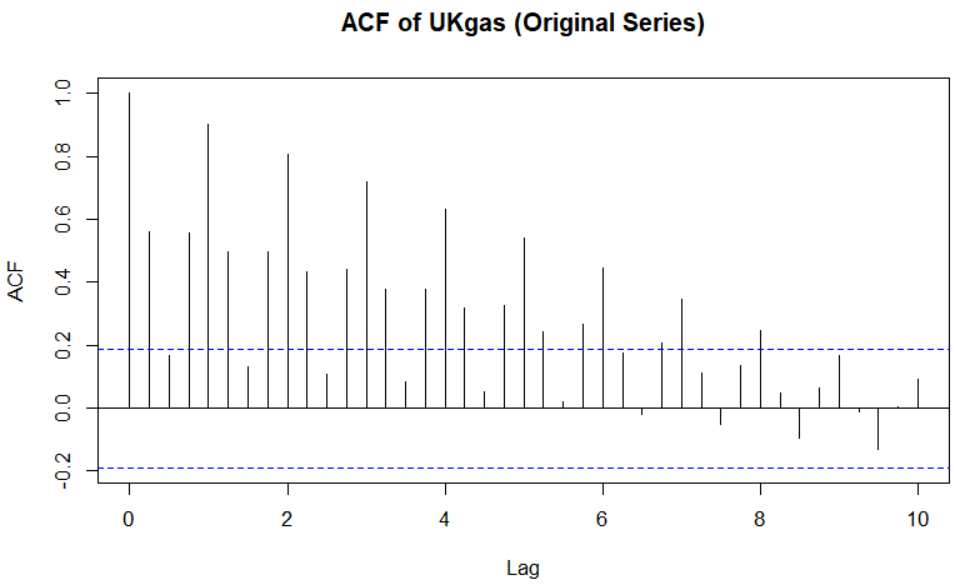


Figure 4: ACF of UKgas (original series)

Figure 4 shows that the ACF decays very slowly and remains well above the significance bands for many lags. This is a classic sign of non-stationarity. A stationary series would have an ACF that drops to near zero relatively quickly. Beyond the slow decay, there are clear spikes at lags 1, 2, 3, and 4 (corresponding to 1, 2, 3, and 4 quarters), with the spike at lag 4 being particularly prominent. This repeating pattern at multiples of 4 confirms the strong quarterly seasonality we already observed visually.

The PACF in Figure 5 tells a different story. There is a very large spike at lag 1 (one quarter), a second notable spike at lag 2, and then the values drop off sharply. This pattern is consistent with an AR structure in the series, but it is important not to read too much into the PACF at this stage. The raw series still contains trend and seasonal non-stationarity, which distorts the PACF. Trying to identify an AR order directly from these plots would

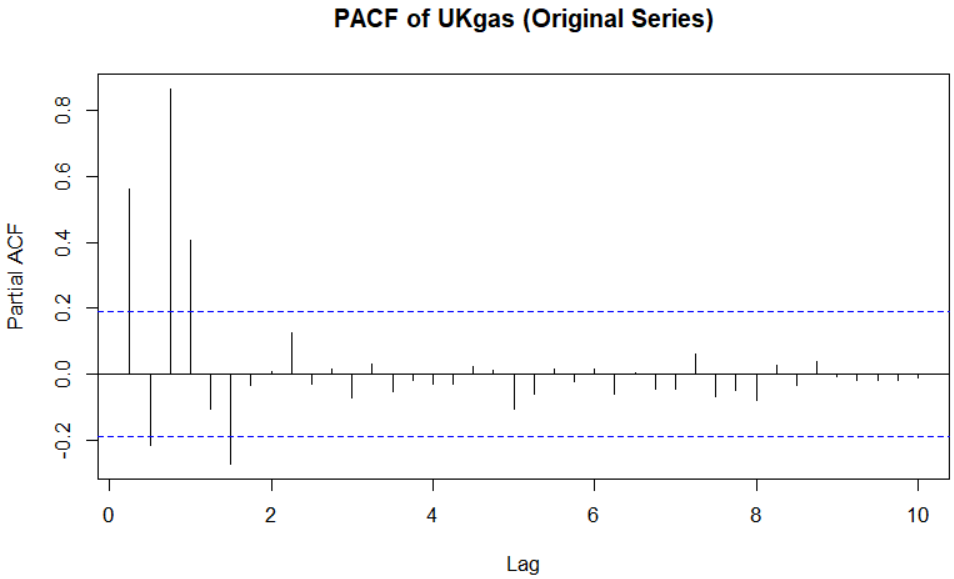


Figure 5: PACF of UKgas (original series)

be premature. The correct approach is to first transform the series toward stationarity and then examine the ACF and PACF of the transformed series for model identification. That is done in Section 6.

Taken together, these plots confirm that the raw UKgas series is not stationary and should not be taken directly into AR, MA, or ARMA fitting.

4 Holt-Winters Forecasting

Since UKgas has both a clear upward trend and strong multiplicative seasonality, the multiplicative form of Holt-Winters exponential smoothing is the appropriate choice here. The additive form would only make sense if the seasonal swings stayed roughly constant over time, which they clearly do not. The multiplicative Holt-Winters model updates three components at each time step: the level, the trend, and the seasonal factors. The smoothing parameters control how quickly each component responds to new observations.

The fitted model produced the following smoothing parameters:

- $\alpha = 0.024$: the level is updated very slowly, meaning the model relies heavily on past observations to estimate the current level rather than reacting to recent changes.
- $\beta = 1.000$: the trend is updated as fast as possible at every step, meaning the model fully incorporates the most recent change in level into its trend estimate. This is unusual and worth noting.
- $\gamma = 0.783$: the seasonal factors update fairly quickly, which gives substantial weight to recent seasonal observations.

The $\beta = 1$ result deserves attention. It means the estimated trend at each point is entirely determined by the most recent change in level, with no smoothing applied. This can make the trend component volatile and sensitive to short-run fluctuations. It does not invalidate the model, but it does suggest the trend in UKgas is not well described by a slowly evolving smooth path, at least within the Holt-Winters framework.

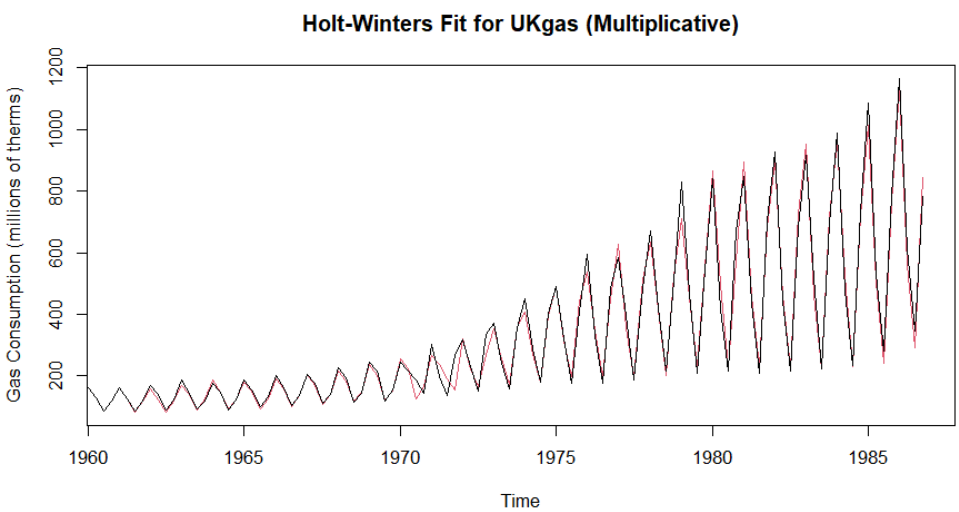


Figure 6: Holt-Winters multiplicative fit for UKgas

Figure 6 shows the fitted values overlaid on the observed series. The model tracks the data closely for most of the sample, capturing both the upward trend and the seasonal pattern well. The fit is tighter in the later part of the sample where the seasonal swings are larger and more distinct.

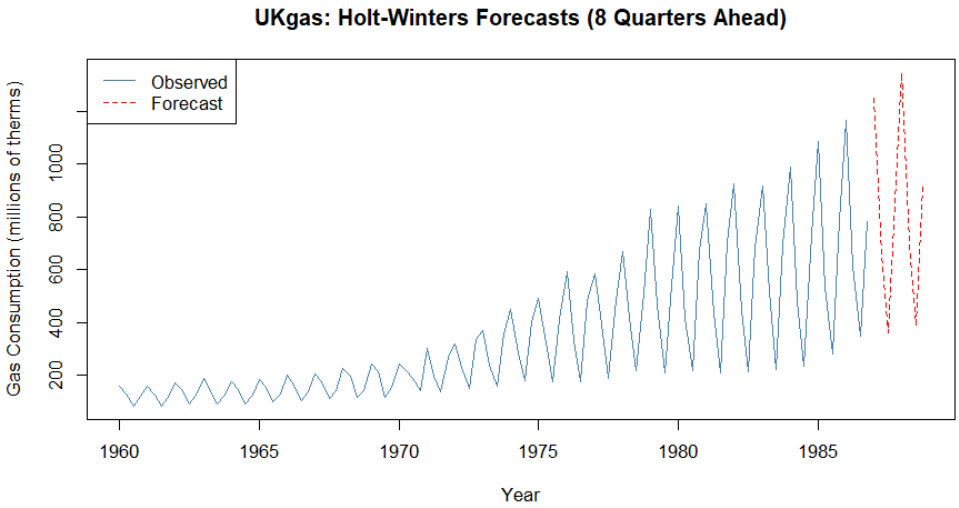


Figure 7: Holt-Winters 8-quarter-ahead forecast for UKgas

Figure 7 shows the 8-quarter-ahead forecast extending beyond 1986. The forecast continues the upward trend and reproduces the seasonal cycle, with Q1 projected to be the highest quarter and Q3 the lowest. This is a sensible and intuitive projection given the structure of the data. However, Holt-Winters is a deterministic smoothing method. It does not model the stochastic dependence structure of the series, and it gives no formal way to assess whether residual autocorrelation remains after the trend and seasonality are accounted for. That question is addressed in the sections that follow.

5 Regression with Trend and Seasonal Dummies

My goal here is to see how much of the systematic structure in UKgas can be explained by a deterministic regression model. Since the raw series has growing variance, we first apply a log transformation to stabilize it before fitting the regression.

Figure 8 shows the log-transformed series. The seasonal swings are now more uniform in size across the sample compared to the raw series, which confirms that the log transformation has done its job of stabilizing the variance. The upward trend and quarterly seasonality are still clearly present, but the amplitude of the seasonal pattern no longer grows as dramatically

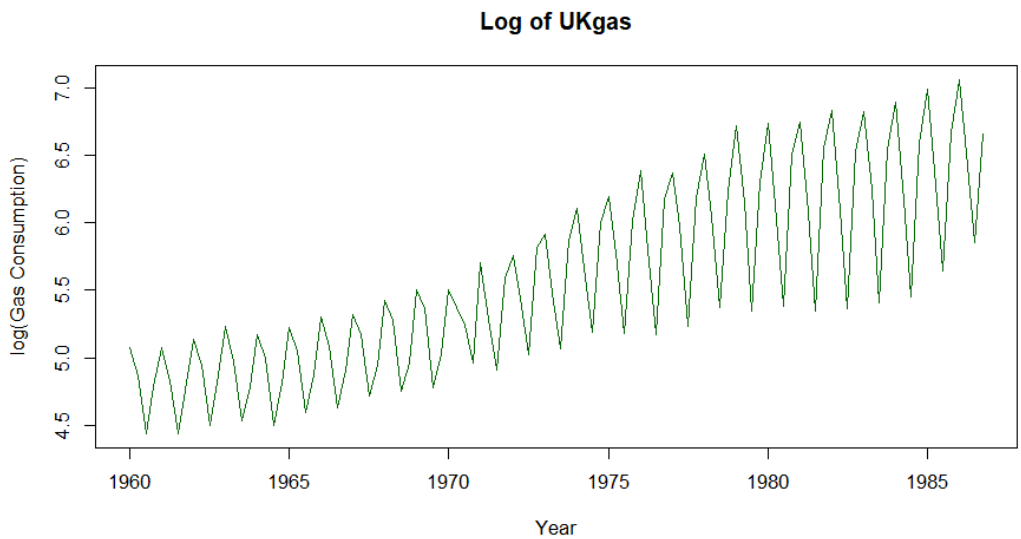


Figure 8: Log-transformed UKgas series

over time. We then fit the following regression model to the log-transformed series:

$$\log(x_t) = \beta_0 \cdot t + \beta_1 Q_1 + \beta_2 Q_2 + \beta_3 Q_3 + \beta_4 Q_4 + \varepsilon_t$$

where t is a numeric time index, and Q_1 through Q_4 are quarterly dummy variables. The intercept is suppressed so that all four seasonal dummies are estimated directly, each representing the average log-consumption level for that quarter after accounting for the time trend. The fitted model gives the following coefficients, all of which are highly statistically significant:

Term	Estimate	Std. Error	t value	p-value	
Time	0.07237	0.00231	31.34	<2e-16	***
Seas1	−136.8	4.556	−30.03	<2e-16	***
Seas2	−137.2	4.556	−30.11	<2e-16	***
Seas3	−137.8	4.557	−30.24	<2e-16	***
Seas4	−137.1	4.557	−30.09	<2e-16	***

*** $p < 0.001$

Table 1: Regression coefficients from the log-linear model with seasonal dummies. Residual standard error: 0.1869 on 103 degrees of freedom. $R^2 = 0.9989$, Adjusted $R^2 = 0.9989$. F-statistic: 1.952×10^4 on 5 and 103 DF, $p < 2.2 \times 10^{-16}$.

The Time coefficient of 0.0724 means that log-consumption increases by approximately 0.072 per unit of the time index per quarter. Since the time variable increments by 0.25 each quarter, the implied quarterly growth rate on the original scale is $e^{0.07237\times 0.25} - 1 \approx 1.8\%$, or roughly 7.5% per year. The seasonal coefficients are large in magnitude but close to each other because the intercept is absorbed into them. What matters is the difference between them: Seas3 is the most negative at -137.8, confirming that Q3 has the lowest average log-consumption, while Seas1 is the least negative at -136.8, confirming Q1 as the highest. The model achieves an R^2 of 0.999, which looks impressive but is partly a consequence of suppressing the intercept and should not be over-interpreted.

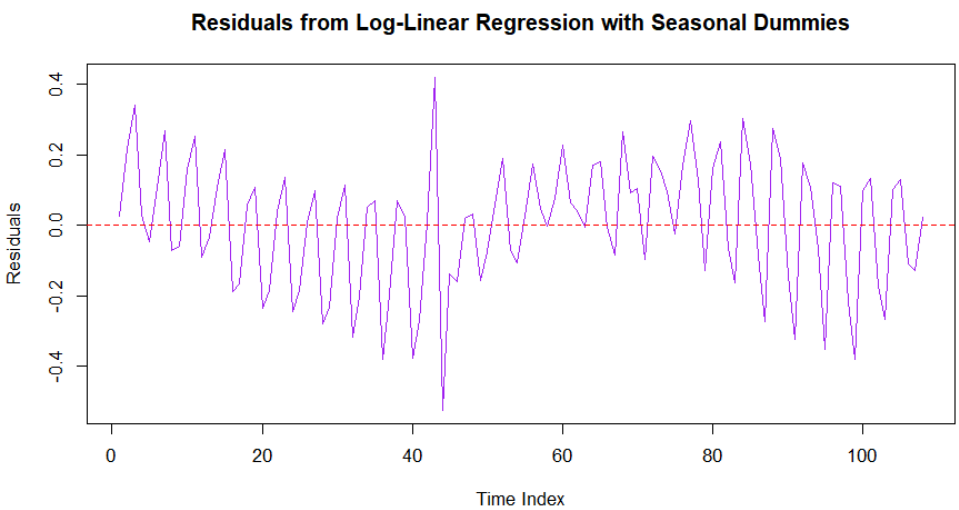


Figure 9: Residuals from the log-linear regression with seasonal dummies

Figure 9 shows the residuals over time. They are not random. There is a visible oscillating pattern that repeats every four observations, which means the regression has not fully removed the seasonal structure. Figure 10 confirms this: there are large spikes in the ACF at lags 4, 8, 12, and 16, all well above the significance bands. This is a strong sign of remaining seasonal autocorrelation in the residuals.

The reason for this is that the regression uses a linear time trend combined with fixed seasonal dummies. This specification assumes that seasonality adds a constant amount to

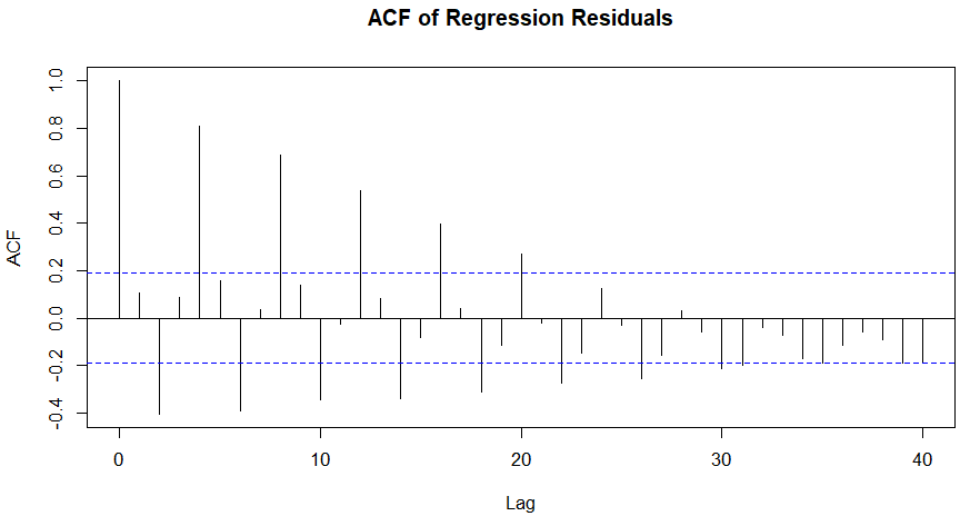


Figure 10: ACF of regression residuals

log-consumption each quarter regardless of where we are in the sample. But even on the log scale, the seasonal pattern in UKgas is not perfectly stabilized, and the deterministic model cannot capture the remaining stochastic dependence. The residual autocorrelation means the OLS assumption of independent errors is violated, and any inference from this model alone would be unreliable.

Going back to Figure 9, we can see a large spike and sudden drop in the residuals around time index 43 (corresponding to late 1970 and early 1971). This is a historical disruption in UK gas supply. Following the discovery of natural gas reserves in the North Sea in the mid-1960s, the UK government launched a nationwide programme to convert the entire gas network from manufactured town gas (produced by heating coal) to natural North Sea gas. This conversion programme ran from 1968 to around 1977 and required converting approximately 40 million appliances across 14 million homes and businesses (National Gas; Resilience). During the peak of this transition in the early 1970s, consumption patterns were temporarily disrupted as regions were switched over one by one, causing short-term irregularities in measured gas demand (Office for Budget Responsibility). The regression model, which assumes a smooth deterministic trend, cannot capture this kind of one-time

structural disruption, so it shows up as a large outlier in the residuals. This is also visible in the random component of the decomposition plot in Section 2.

6 Transformation Toward Stationarity

The raw UKgas series cannot be used directly for stationary AR, MA, or ARMA modeling because it has a clear upward trend and strong quarterly seasonality. Both of these features violate the stationarity assumption, which requires the mean and autocorrelation structure of the series to be constant over time. Therefore, we need to apply two transformations to bring the series closer to stationarity.

Step 1: Log Transformation

The log transformation was already applied in Section 5 to stabilize the growing variance. Since the seasonal swings in the raw series grow proportionally with the level, taking the log compresses the scale and makes the variance more uniform across the sample. This is a necessary first step before differencing.

Step 2: Seasonal Differencing

After logging, the series still has a trend and seasonal structure. We apply a seasonal difference at lag 4, which subtracts from each observation the value from the same quarter one year earlier:

$$y_t = \log(x_t) - \log(x_{t-4})$$

This removes the quarterly seasonal pattern and much of the trend in one step.

Figure 11 shows the seasonally differenced log series. The series now fluctuates around zero with no obvious upward trend and no repeating seasonal pattern, which is a good sign. There is one prominent disruption around 1970–1971, which again reflects the town gas to natural

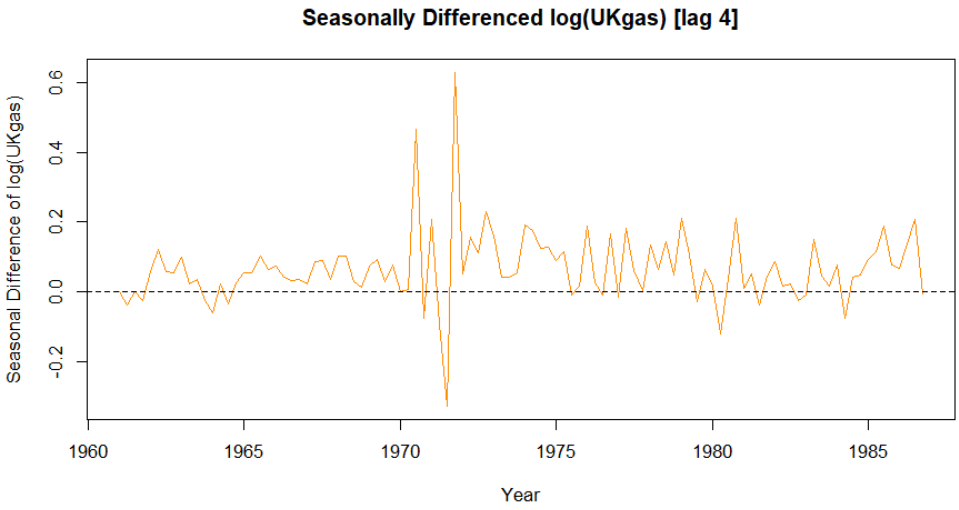


Figure 11: Seasonally differenced log(UKgas) at lag 4

gas conversion discussed in Section 5. Apart from that episode, the series looks reasonably stable.

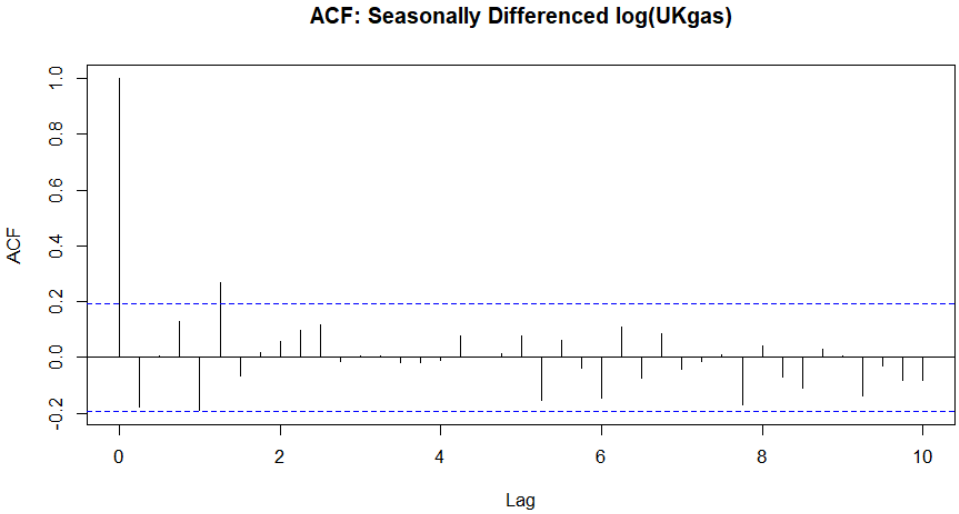


Figure 12: ACF of the seasonally differenced log(UKgas)

The ACF in Figure 12 shows only a couple of significant spikes at the first few lags and then drops within the significance bands for most subsequent lags. There are no large spikes at seasonal lags 4, 8, or 12, which confirms that the seasonal differencing has successfully removed the quarterly structure.

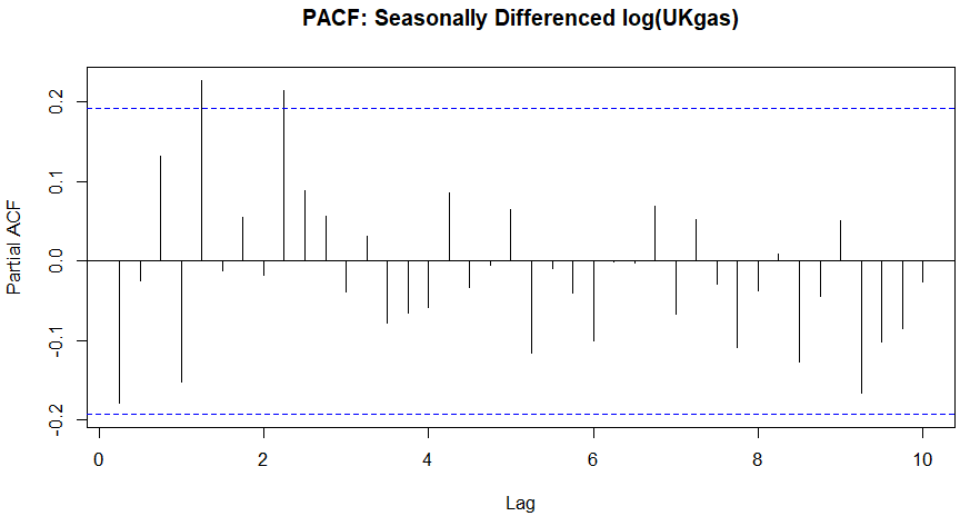


Figure 13: PACF of the seasonally differenced log(UKgas)

The PACF in Figure 13 also shows a small number of significant lags at the beginning and then tails off. This pattern is consistent with a low-order stationary process, which means the transformed series is now a reasonable candidate for AR, MA, or ARMA fitting.

Double Differencing: Why We Do Not Use It

As an additional check, a first difference was applied on top of the seasonal difference to see whether further differencing improves stationarity.

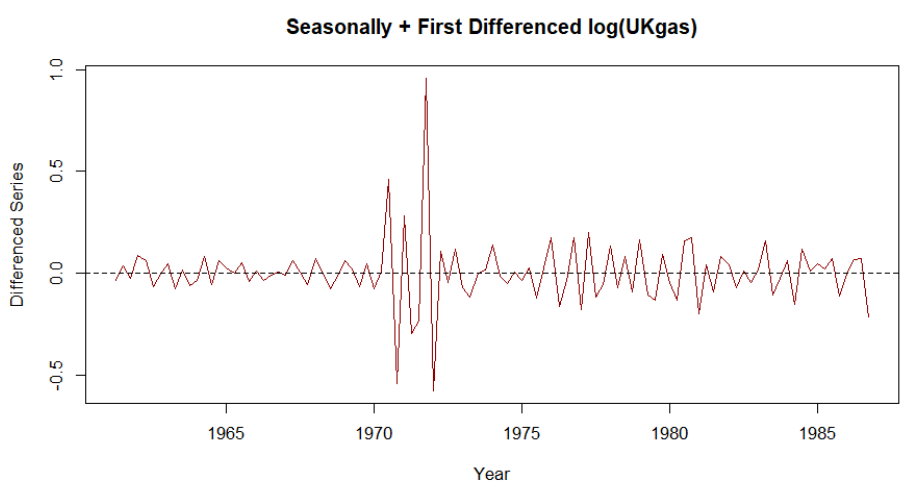


Figure 14: Seasonally and first differenced log(UKgas)

Figure 14 shows that double differencing amplifies the 1970–1971 disruption dramatically, producing extreme spikes that dominate the entire plot. The ACF in Figure 15 also becomes harder to interpret, with more significant lags rather than fewer. This is a sign of over-differencing: applying more differences than necessary does not improve stationarity, it introduces new structure into the series. For this reason, the seasonally differenced log series is used as the working stationary series for all model fitting.

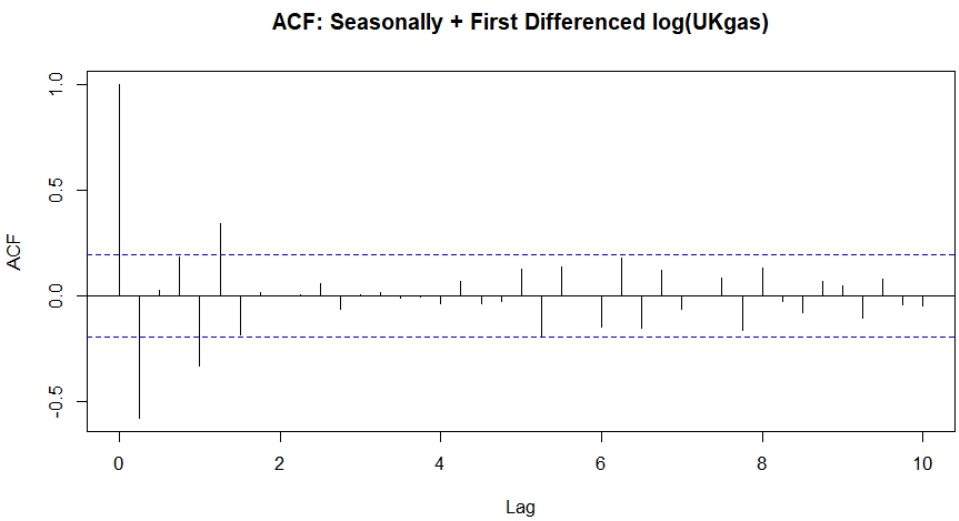


Figure 15: ACF of the seasonally and first differenced log(UKgas)

7 AR, MA, and ARMA Modeling

With the seasonally differenced log series established as approximately stationary, we now fit three candidate models and compare them formally. The three models considered are AR(1), MA(1), and ARMA(1,1). The goal is not to assume any one of these is correct in advance, but to let the data and formal criteria guide the choice.

An AR(1) model says that the current value of the series depends on its own value one period ago plus a white noise error:

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$

An MA(1) model says the current value depends on the current error and the error from one period ago:

$$y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

An ARMA(1,1) model combines both:

$$y_t = \phi_1 y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

All three models are fitted to the stationary series using maximum likelihood estimation, and compared using the Akaike Information Criterion (AIC). AIC penalizes model complexity, so a lower AIC indicates a better balance between fit and parsimony.

Model	Log-Likelihood	Parameters	AIC
AR(1)	87.76	3	−169.52
MA(1)	87.73	3	−169.46
ARMA(1,1)	87.77	4	−167.55

Table 2: AIC comparison of candidate stationary models

Table 2 shows the results. AR(1) has the lowest AIC at −169.52, followed very closely by MA(1) at −169.46. The difference between these two is only 0.06, which is negligible and means both models fit the transformed series almost equally well. ARMA(1,1) has the highest AIC at −167.55, which means adding both an AR and an MA term together does not improve the model enough to justify the extra parameter. This is a case where a simpler model wins.

The ARMA(1,1) result is worth examining more carefully. The estimated coefficients are $\hat{\phi}_1 = -0.121$ with a standard error of 0.374, and $\hat{\theta}_1 = -0.060$ with a standard error of 0.369. Neither coefficient is statistically significant, which means the model cannot reliably distinguish the AR and MA components from each other. This is a common problem when both terms are included unnecessarily, and it gives another reason to prefer the simpler models.

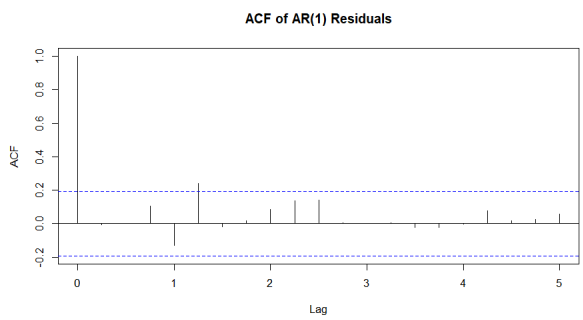
The fitted AR(1) model is:

$$\hat{y}_t = -0.178\, y_{t-1} + 0.0658, \quad \hat{\sigma}^2 = 0.01083$$

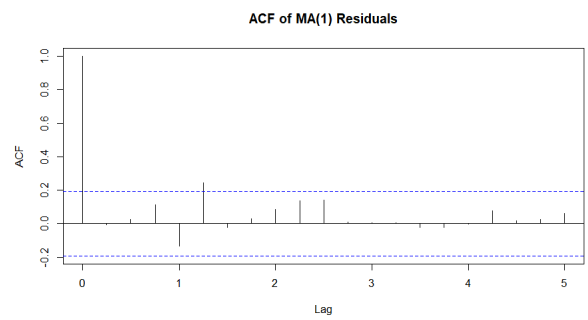
The fitted MA(1) model is:

$$\hat{y}_t = -0.172\, \varepsilon_{t-1} + 0.0658, \quad \hat{\sigma}^2 = 0.01083$$

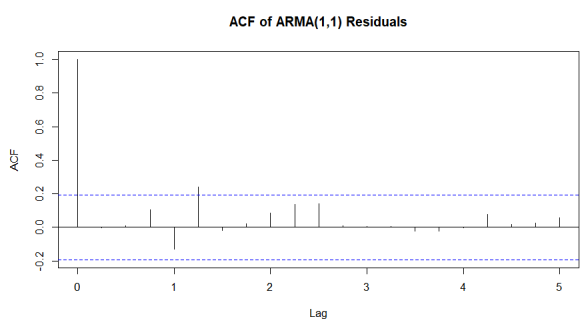
Both models estimate essentially the same intercept and the same residual variance, which further confirms how similar they are in practice.



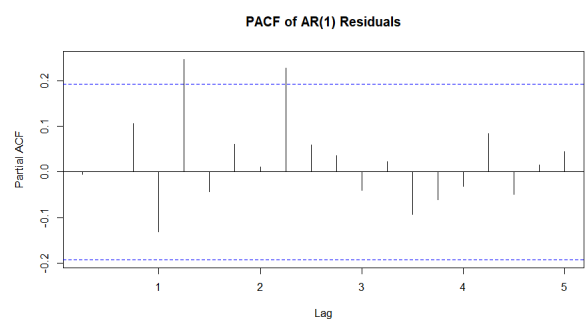
(a) ACF of AR(1) residuals



(b) ACF of MA(1) residuals



(c) ACF of ARMA(1,1) residuals



(d) PACF of AR(1) residuals

Figure 16: Residual diagnostics for the three candidate models

Figures 16a, 16b, and 16c show the residual ACFs for all three models. All three plots look nearly identical, with almost all lags falling within the significance bands. This confirms that all three models adequately capture the dependence structure of the transformed series, and

that the choice between them cannot be made on residual diagnostics alone. The PACF of the AR(1) residuals in Figure 16d also shows no significant structure remaining.

Based on AIC and the insignificance of the ARMA(1,1) coefficients, AR(1) is selected as the preferred model. MA(1) would be an equally defensible choice given how close the two models are, but AR(1) has the marginally lower AIC and is slightly easier to interpret in terms of how past values of the series feed into current values.

8 Forecasting with the Preferred Stochastic Model

The preferred model from the analysis is AR(1), fitted to the seasonally differenced log series. We generate forecasts 8 steps ahead, covering Q1 1987 through Q4 1988. Table 3 reports the forecast values and their standard errors on the transformed scale.

Quarter	Forecast	Std. Error
Q1 1987	0.0786	0.1040
Q2 1987	0.0635	0.1057
Q3 1987	0.0662	0.1057
Q4 1987	0.0657	0.1057
Q1 1988	0.0658	0.1057
Q2 1988	0.0658	0.1057
Q3 1988	0.0658	0.1057
Q4 1988	0.0658	0.1057

Table 3: AR(1) forecasts on the seasonally differenced log scale

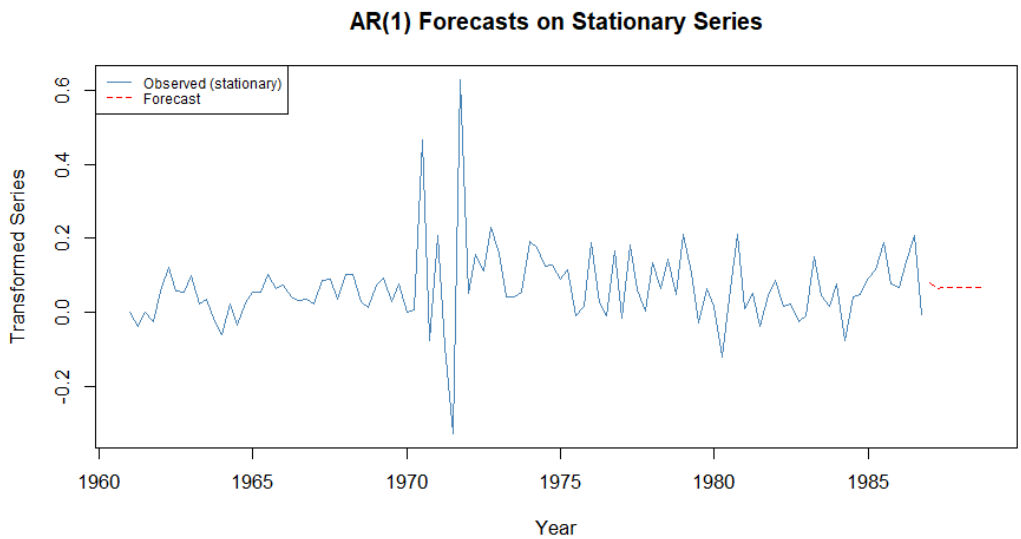


Figure 17: AR(1) forecasts appended to the stationary series

Figure 17 shows the stationary series with the 8-step forecast appended as a dashed red line. The forecast quickly converges to a flat line near 0.066, which is the long-run mean of the AR(1) process. This is expected behavior: because the AR(1) coefficient is small ($\hat{\phi}_1 = -0.178$), the series has weak memory and the forecast reverts to the mean after just a couple of steps. The standard errors are around 0.105 for all steps beyond the first, which reflects the uncertainty that accumulates when forecasting further ahead.

It is important to note that these forecasts are on the transformed scale, that is, the seasonally differenced logarithm of gas consumption. They cannot be read directly as gas consumption figures. To convert them back to the original scale would require reversing both the seasonal differencing and the log transformation, which involves adding back the log values from four quarters earlier and then exponentiating. This back-transformation was not performed here because the primary purpose of the AR(1) model is to capture the stochastic dependence structure of the stationary series, not to produce a standalone forecast on the original scale.

For a forecast on the original scale, the Holt-Winters method from Section 4 is more directly useful. That model produced forecasts in units of millions of therms, continuing the upward

trend and seasonal cycle in an intuitive way. The two approaches serve different purposes: Holt-Winters is better for producing interpretable point forecasts on the original scale, while the AR(1) model provides a statistically grounded description of the residual dependence after trend and seasonality have been removed.

9 Conclusion

The UKgas series has a strong upward trend, sharp quarterly seasonality, and seasonal swings that grow with the level. These three features forced every methodological choice in the paper: multiplicative decomposition over additive, Holt-Winters in its multiplicative form, a log transformation before regression, and seasonal differencing at lag 4 to reach stationarity.

What this analysis ultimately shows is that the same data needs different tools for different questions. Holt-Winters gave the most useful forecasts on the original scale, but it could not say anything about whether stochastic dependence remained. The regression model captured the broad structure but left clear residual autocorrelation, partly because a linear trend with fixed dummies cannot match a multiplicative seasonal pattern and partly because of the 1970–1971 disruption from the UK's town gas to natural gas conversion. Only after transformation could AR(1) be fitted, and even then the choice between AR(1) and MA(1) came down to a tiny AIC difference. ARMA(1,1) was rejected because the extra parameter was not statistically distinguishable from zero.

The takeaway is that more complexity is not always better. AR(1) won here because the data did not need anything more, and recognizing that is just as much a result as the model itself.

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